

- 3 (a) Use the method of differences to find $\sum_{r=1}^n \frac{1}{(kr+1)(kr-k+1)}$ in terms of n and k , where k is a positive constant. [4]

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(b) Deduce the value of $\sum_{r=1}^{\infty} \frac{1}{(kr+1)(kr-k+1)}$. [1]

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(c) Find also $\sum_{r=n}^{n^2} \frac{1}{(kr+1)(kr-k+1)}$ in terms of n and k . [2]

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